



Panaroial Mark

Spending millions on his security, Mark Zuckerberg still chooses to tape his webcam, audio jack and microphones. <http://dgit.in/ParanoMark>



Couponhaat's app

Couponhaat's new smartphone app on Android and iOS lets you access discounts. <http://dgit.in/CHtApp>

GPU comparison

excitement in the GPU arena than now. It'll be at least four or five months before we see AMD and NVIDIA's flagships but till then we'll have to settle for the GTX 1070 and the RX 480. (Well, not settle exactly.)

Then there's the fact that both NVIDIA and AMD have moved to smaller transistor

sizes, NVIDIA with its 16nm FinFET and AMD going with 14nm FinFET. NVIDIA has been moving down the nanometer scale on a regular basis but this Polaris was AMD's transition after a really long time on 28nm. This has great implications on the power efficiency as smaller transis-

tors can operate at lower voltages and also generates a lot less heat. This means that AMD cards might finally shrug off the hot-head label but that's to be confirmed once we get the test sample.

Apparently, HDR is really blowing up. Both NVIDIA and AMD are really pushing

How We Tested

We test graphics cards for gaming performance, compute performance and then assess their build quality. With these three parameters quantified we then take the price into consideration to determine the performance per rupee value. India being a price conscious market is quite keen on getting more for the money. Another reason why value for money matters is we endure a lot of taxation and price-gouging, so the price-to-performance ratio is quite different than what these cards were initially priced at by their respective companies. And lastly, we check the build quality of each graphics card. With that done, we formulate scores and decide upon the winners across the categories. Read on to get a more detailed explanation of the benchmarks used.

Here's the configuration we went with for the test:

- **Processor:** Intel Core i7 4960X
- **Motherboard:** ASRock X79 Extreme 11
- **RAM:** 4x 8 GB Corsair Dominator Platinum CMD32GX3M4A2133C9
- **Boot drive:** 2x SanDisk Extreme II 240 GB SSD
- **Cooler:** Corsair H110i GT Liquid Cooler
- **SMPS:** Antec HCP-1000 Platinum
- **Monitor:** BenQ BL3201PT

Performance benchmarks

Synthetic

3DMark

No graphics card benchmark is ever complete without a few runs of 3DMark. The all rounder has multiple benchmarks for testing graphics cards from all segments, from flagships to entry-level kiddies. However, VR isn't 3DMark's strong suit owing to the fact that it doesn't take the headset into consideration. So we don't perform VR tests using 3DMark.

FurMark

Part of the OpenGL workload, this benchmark is

often called a power-virus which is not configured properly before running can cause your graphics card to heat up and go kaput. We use this to record peak load temperatures and power consumption, for obvious reasons.

LuxMark

LuxMark uses OpenCL framework to render a 3D scene and helps us benchmark a cards parallel processing abilities. Each run takes 120 seconds and the maximum number of frames rendered is considered the score.

API Overhead Feature Test

Still in the nascent stage, 3DMark's API overhead feature test benchmarks the GPU's ability to handle DirectX11, 12 and Mantle's API calls per unit time. Mantle may no longer be on the horizon but it gave rise to Vulkan which is the next big 3D graphics and compute API. We do not consider the scores towards our final calculations.

SteamVR

Steam's own VR Performance Test is a good way for you to allay your doubts over your graphics card's ability to handle VR devices like the Oculus or HTC's Vive.

Games

We've been testing graphics cards with a bunch of games and we retain our older game benchmarks just so that we can compare graphics cards across generations. The following games are part of our legacy suite.

- Bioshock Infinite
- Tomb Raider
- Sleeping Dogs
- Hitman Absolution

And then we have our updated suite which consists of the following games:

- DOOM
- Ashes of the Singularity

- Total War: Warhammer
- Witcher 3
- Rise of the Tomb Raider
- Fallout 4
- GTA V

Given the fact that we have limited space in the magazine, we only include one of these games for showing FPS scores. The remaining FPS scores can be found on the individual reviews when they're published online.

Build Quality

With changes in the GPU microarchitecture, the performance difference caused by a few MHz here and there don't matter. Even minor overlocks rarely show a difference in performance. So manufacturers have increasingly started focussing on improving the build quality of the graphics card to justify higher prices. The cooler is of prime focus here. Having a large enough heat sink, durable and silent fans bearings, improved fan fins, vapour chambers, etc., all contribute towards a better graphics card. Some manufacturers take the extra pain to add RGB which we consider purely cosmetic but there are those who modify the PCB to add more power phases to enable a precise current flow for overclocking. Those are taken into consideration as well.

Value for money

The total performance score generated by a particular card is divided by the cost of the card to deduce the performance/rupee factor. Often this leads to the most cheapest card in the lot being picked as the card that offers the most value for your money. However, this parameter also helps decide between close competitions as cards with the same GPU often end up being very close in terms of performance. This is where the VFM comes into the picture and decides the winner.